

Manish Kumar

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Professional Summary

Machine Learning Systems & Architecture Engineer with a proven track record of training generative foundation models on large-scale distributed infrastructure. Deep expertise in designing and optimizing complex mathematical architectures, including Denoising Diffusion Probabilistic Models (DDPMs) and Space-Time U-Nets, on high-performance computing (HPC) clusters. Adept at scaling training loops, managing node-to-node communication bottlenecks, and translating cutting-edge AI research into highly optimized, production-ready distributed systems.

Technical Skills

- **Core Architectures:** Generative Foundation Models, Denoising Diffusion Probabilistic Models (DDPM), Diffusion Transformers (DiT), Spatio-Temporal Modeling, Space-Time U-Nets.
- **Distributed Infrastructure & Compute:** Multi-Node Distributed Training, High-Performance Computing (HPC), AIRAWAT Supercomputer facility, Cluster Optimization (Slurm, NCCL), VRAM Memory Management.
- **Frameworks & Tools:** PyTorch, Hugging Face Diffusers, Accelerate, DeepSpeed, Transformers, Python, Linux/Bash, xarray, WRF.
- **Domain Expertise:** Applied Mathematics, Algorithm Optimization, Parallel Computing, Large-Scale Data Assimilation, Mathematical Modeling.

Professional Experience

Zulense Private Limited | Gurugram, India *Founder & Lead AI Architect* | December 2025 – Present

- **Foundation Model Engineering:** Architected and trained the *Z1 Imagination Engine*, implementing custom Denoising Diffusion Probabilistic Models (DDPM) with spatio-temporal U-Net backbones to handle high-dimensional video and physical simulation generation.
- **Supercomputing Scaling:** Directed end-to-end distributed training pipelines on the national **AIRAWAT supercomputer facility**. Optimized multi-node GPU utilization,

resolved distributed data-loading bottlenecks, and managed large-scale cluster orchestration to reduce training wall-clock time.

- **Algorithmic Optimization:** Designed a highly scalable cognitive architecture ("Teacher Brain") leveraging structured data-assimilation pipelines and complex logic trees to systematically automate conceptual generation workflows aligned with national educational frameworks.
- **Technical Leadership:** Recruited and led an elite research team comprising M.Tech and Ph.D. talent from tier-1 institutions (IITs), driving the transition from theoretical mathematical proofs to production-level distributed code. Secured seed funding by demonstrating viable execution of core AI pipelines.

Education

National Institute of Technology (NIT) Patna | *Patna, Bihar, India B.Tech in Applied Mathematics and Computer Science | Graduated: 2023*

- **Focus Areas:** Numerical Methods, Stochastic Processes, Linear Algebra, Mathematical Modeling, and Climate Science (WRF).